**DBMS – SEPHORA BEAUTY SALES AND CUSTOMER ANALYTICS SYSTEM**

**WEEK 9**

1. AGGREGATE FUNCTIONS

SQL Aggregate Functions are used to perform calculations on a group of rows and return a single result. Sephora Beauty uses Aggregate Functions to analyse its online beauty product sales by calculating the total orders, total sales revenue, average order value, and the highest and lowest order amounts.

### 1.1 COUNT() – Total Number of Orders

```
SELECT COUNT(*) AS Total_Orders
FROM SephoraOrder;
```

*[Paste output screenshot here]*

### 1.2 SUM() – Total Sales Amount

```
SELECT SUM(Total_Amount) AS Total_Sales
FROM SephoraOrder;
```

*[Paste output screenshot here]*

### 1.3 AVG() – Average Order Amount

```
SELECT AVG(Total_Amount) AS Average_Order_Amount
FROM SephoraOrder;
```

*[Paste output screenshot here]*

### 1.4 MIN() – Minimum Order Amount

```
SELECT MIN(Total_Amount) AS Minimum_Order_Amount
FROM SephoraOrder;
```

*[Paste output screenshot here]*

### 1.5 MAX() – Maximum Order Amount

```
SELECT MAX(Total_Amount) AS Maximum_Order_Amount
FROM SephoraOrder;
```

*[Paste output screenshot here]*

### 2. GENERATE TOTAL SALES REPORT

```
SELECT
    COUNT(Order_ID) AS Total_Orders,
```

```
    SUM(Total_Amount) AS Total_Sales,
    AVG(Total_Amount) AS Average_Sales
FROM SephoraOrder;
```

*[Paste output screenshot here]*

**2.1 Customer-wise Sales Report**

```
SELECT
    c.Customer_ID AS Customer_ID,
    c.Customer_Name AS Customer_Name,
    SUM(o.Total_Amount) AS Total_Purchase_Amount
FROM SephoraCustomer c
JOIN SephoraOrder o
    ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Purchase_Amount DESC;
```

*[Paste output screenshot here]*

**3. Top Customers Based on Purchase Amount**

```
SELECT
    c.Customer_ID AS Customer_ID,
    c.Customer_Name AS Customer_Name,
    SUM(o.Total_Amount) AS Total_Purchase
FROM SephoraCustomer c
JOIN SephoraOrder o
    ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Purchase DESC
FETCH FIRST 5 ROWS ONLY;
```

*[Paste output screenshot here]*

**4. Best-Selling Products**

```
SELECT
    p.Product_ID,
    p.Product_Name,
    SUM(oi.Quantity) AS Total_Quantity_Sold
FROM SephoraProduct p
JOIN SephoraOrder_Item oi
    ON p.Product_ID = oi.Product_ID
GROUP BY p.Product_ID, p.Product_Name
ORDER BY Total_Quantity_Sold DESC;
```

*[Paste output screenshot here]*

### 4.1 Product Sales – Amount and Quantity

```
SELECT
    p.Product_ID,
    p.Product_Name,
    SUM(oi.Quantity) AS Total_Quantity_Sold,
    SUM(oi.Total_Amount) AS Total_Sales_Amount
FROM SephoraProduct p
JOIN SephoraOrder_Item oi
    ON p.Product_ID = oi.Product_ID
GROUP BY p.Product_ID, p.Product_Name
ORDER BY Total_Sales_Amount DESC;
```

*[Paste output screenshot here]*

### 5. Category-wise Sales Analysis

```
SELECT
    c.Category_ID,
    c.Category_Name,
    SUM(oi.Quantity) AS Total_Quantity_Sold,
    SUM(oi.Total_Amount) AS Total_Sales
FROM SephoraCategory c
JOIN SephoraProduct p
    ON c.Category_ID = p.Category_ID
JOIN SephoraOrder_Item oi
    ON p.Product_ID = oi.Product_ID
GROUP BY c.Category_ID, c.Category_Name
ORDER BY Total_Sales DESC;
```

*[Paste output screenshot here]*

### 6. Combined Business Sales Report

```
SELECT
    c.Customer_ID AS Customer_ID,
    c.Customer_Name AS Customer_Name,
    o.Order_ID,
    TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
    p.Product_Name,
    cat.Category_Name,
    oi.Quantity,
    oi.Total_Amount AS Item_Amount,
    o.Order_Status
FROM SephoraCustomer c
JOIN SephoraOrder o
    ON c.Customer_ID = o.Customer_ID
JOIN SephoraOrder_Item oi
    ON o.Order_ID = oi.Order_ID
```

```
JOIN SephoraProduct p
    ON oi.Product_ID = p.Product_ID
JOIN SephoraCategory cat
    ON p.Category_ID = cat.Category_ID
ORDER BY o.Order_ID;
```

*[Paste output screenshot here]*